NORTH SOUTH UNIVERSITY

Centre of Excellence in Higher Education

DEPARTMENT OF MATHEMATICS AND PHYSICS

School of Engineering and Physical Sciences

Course Code: AMCS 504L
Course Title: **Advanced Numerical Methods & Computation Lab**

Project Problems: Show the steps in your programming code to answer each part

1. Consider the electrostatic potential $u(r)$ between two concentric spheres of Radii $R_1 = 2$ and $R_2 = 4$. The potential of the inner sphere is kept constant at V_1 volts, and the potential of the outer sphere is 0 volts. The potential in the region between the two spheres is governed by,
$$\frac{1}{r^2}\frac{d}{dr}\left[r^2\frac{du}{dr}\right] = 0, \qquad R_1 \le r \le R_2, \qquad u(R_1) = 110, \qquad u(R_2) = 0$$
(a) Approximate $u(r)$ for $r = 2.5, 3.0, 3.5$ using
   (i) Linear-Shooting Method
   (ii) Finite difference Method for linear problems
(b) Compare the accuracy of your results obtained in (a) with the exact solution
$$u(r) = \frac{110\,R_1}{r}\left(\frac{R_2 - r}{R_2 - R_1}\right)$$
(c) Write your comments on the strength of the methods based on your numerical observations.

2. Consider the stiff non-linear IVP that governs the flow of current in a vacuum tube
$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x = 0, \qquad x(0) = 1,\ x'(0) = 0$$
where $\mu > 0$ is a parameter governing nonlinearity.
(a) Convert the IVP into a system of ODEs.
(b) Plot the phase portrait of the solution of the system of ODE for $\mu = 0, 1, 2$
(c) Plot the solution of the ODE for $0 \le t \le 10$.
(d) Write your observation and find a physical interpretation of the solution found above.

3. Consider the problem of predicting the population of two species (x & y) that compete for the same food supply. Assume that the population of a particular pair of species is described by the equations
$$\frac{dx}{dt} = x\,(k_1 - k_2x - k_3y), \qquad \frac{dy}{dt} = y\,(k_4 - k_5x - k_6y)$$
(a) Approximate the population of each species for $0 \le t \le 7$ if $k_1 = 3.0$; $k_2 = 0$; $k_3 = 0.002$; $k_4 = -0.5$; $k_5 = -0.0006$; $k_6 = 0.0$; and the initial population of each species is $x(0) = 1000$, $y(0) = 500$.
(b) Plot the graph of the solution of each species with time and the phase portrait of both species.
(c) Is there a stable solution to this population model? If so, for what values of x_1 and x_2 is the solution stable?

4. Let $P(t)$ be the number of the population at time t, measured in years. If the average birth rate b is constant and the average death rate d is proportional to the size of the population (due to overcrowding), then the growth rate of the population is modeled by the logistic equation
$$\frac{dP}{dt} = (b - d)P \text{ where } d = kP(t).$$
(a) Approximate the population for every year for the next five years if $b = 2.9 \times 10^{-2}, k = 1.4 \times 10^{-7}$ with the initial population 50976.
(b) Find the accuracy of your results with the exact solution by solving the above problem analytically.